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Flame characteristics of turbulent lean premixed methane/air flames at high pressure: Turbulent flame speed and flame brush thickness

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Abstract

The effects of operating conditions and turbulence on flame front position, turbulent flame speed and flame brush thickness of lean premixed methane/air flames at high pressure are investigated experimentally. A comparison of the measured turbulent flame speeds with existing correlations is presented and discussed. The measurements were performed in an axis-symmetric, generic combustor at pressures up to $p = 1.44$ MPa, equivalence ratios in the range $\Phi = 0.43$–0.56, preheating temperatures $T = 673$–773 K and for an inlet bulk velocity range of $u_{\mathrm{BULK}} = 30$–60 m/s. The turbulence intensity and the integral length scale at the combustor inlet were varied by means of turbulence grids with different geometry and by changing the grid position in the inlet channel. The isothermal flow field was characterized by two-dimensional particle image velocimetry (PIV). The characterization of the flame front was based on the statistical analysis of two-dimensional instantaneous images of the laser induced fluorescence of the OH radical (OH-PLIF). The analyses revealed that pressure has no effect on the flame front position, the turbulent flame speed, or on the flame front fluctuation (flame brush thickness). Decreasing the fuel concentration from $\Phi = 0.43$ to $\Phi = 0.56$ results in a flame elongation by a factor of 2 and a corresponding decrease of the turbulent flame speed by a factor of 3. For the same variation of the fuel concentration the flame brush thickness increases by a factor of 2. Additionally, the flame brush thickness was consistently observed to vary proportionally to the square root of the flame front position. The experimental values of the turbulent flame speed are in satisfactory agreement with the existing correlations.
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Keywords: High-pressure lean premixed flames; Turbulent flame speed; Flame brush thickness

1. Introduction

Currently, lean premixed combustion is the favored method for low-emission power generation from natural gas used in stationary gas turbines. Homogenous mixing of the fuel and air combined with ultra-lean operation provide a reduction of flame temperature and are key to its success in minimizing NO_x formation. Increasing environmental concerns will lead to heavier restrictions on NO_x emissions from gas turbines prompting a need to exploit this combustion technique to its maximum potential. Further reduction of NO_x will require extremely lean mixtures, therefore a better understanding of the turbulence-chemistry interaction is needed which

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remains one of the great challenges in combustion research [1]. A general approach in combustion research has been the characterization of turbulent premixed flames in terms of the turbulent flame speed s_T and the thickness δ_T. The experimental investigation of the turbulent flame speed and thickness (also referred to as mean flame brush thickness) can help to validate numerical models [2] such as the model proposed by Zimont [3]. This model is based on the infinitely-fast-chemistry approach following the well-known Bray–Moss–Libby model [4] for the turbulence flame speed closure. The turbulent flame speed and flame front fluctuation are also of prime importance for the combustor design [5,6]. In particular, a better understanding of the effects of pressure and turbulence on s_T and δ_T is needed.

Turbulent flame speed data from current literature are quite scattered and attempts to derive a universal empirical correlation that can predict s_T for a broad range of thermo-chemical and turbulence conditions have not been successful [2,7]. For flames typical for gas turbines there are, however, several correlations that have gained a relative popularity [7,8]. The agreement between different experimental data remains unsatisfactory because the turbulent flame speed does not only depend on turbulence and the operating conditions but also strongly depends on the determination method and, in particular, depends on the flame configuration [1].

The majority of the literature data dealing with turbulent premixed flames are not representative for gas turbine conditions. The work of Abdel-Gayed et al. [9] comprises a large data set measured in a fan-stirred bomb but only for atmospheric conditions. Turbulent flame speeds at atmospheric pressure were also measured by Plessing et al. [10] in a low-swirl burner. Recent measurements of Bradley et al. [11] were performed in a fan-stirred bomb up to 0.5 MPa and different gaseous fuels but for methane only at stoichiometric mixtures. The influence of pressure on the flame front curvature was studied by Soika et al. [12] for pressures up to 1.1 MPa and equivalence ratios between $\Phi = 0.7$–1.1. Because of the limited range of equivalence ratios these results are also not completely representative for gas turbine conditions. Recent work of Kobayashi et al. [13] comprises measurements of turbulent flame speeds in a Bunsen-type burner for pressures up to 1 MPa. These flames belong to the corrugated flamelets regime rather than to the thin reaction zone regime typical for gas turbines as classified in the Borghi/Peters diagram [14,15].

The lack of experimental data dealing with turbulent premixed flames at gas turbine relevant conditions motivated the current research which aims at the characterization of the turbulent lean premixed methane/air flames in the thin reaction zones regime (or thickened flames regime). These flames are characterized by high turbulent Reynolds numbers ($Re_T > 10^3$), Damköhler numbers lower than 10, and Karlovitz numbers higher than 10 [16]. The characteristic parameters are defined in Eq. (1).

$$Re_T = \frac{u'}{s_L} \cdot \frac{L_{T,xx}}{\delta_L}, \quad Da = \frac{L_{T,xx}}{\delta_L} \cdot \left(\frac{u'}{s_L}\right)^{-1},$$
$$Ka = \left(\frac{u'}{s_L}\right)^{\frac{3}{2}} \cdot \left(\frac{L_{T,xx}}{\delta_L}\right)^{-\frac{1}{2}}. \tag{1}$$

In this work, the laminar flame parameters (flame speed s_L and flame thickness δ_L) were calculated using correlations from [17] and the turbulence intensity u' and the integral length scale $L_{T,xx}$ (based on the correlation of the axial velocity component in the x direction) were measured with PIV.

The flame characteristics were measured at pressures up to 1.44 MPa (absolute), lean equivalence ratios ($\Phi = 0.43$–0.56), preheating temperatures of $T = 673$–773 K and for inlet bulk velocities of $u_{BULK} = 30$–60 m/s. The turbulence intensity and integral length scale at the combustor inlet were varied by means of turbulence grids with different geometry and by changing the grid position. The position and fluctuation of the flame front are characterized by a statistical analysis of OH-PLIF images and their dependence on operating conditions (p, Φ, T, u_{BULK}) and turbulence quantities (u', $L_{T,xx}$) are discussed. Turbulent flame speeds are presented and compared to correlations from literature.

2. Experimental set-up and methodology

The experiments were carried out in a generic, high-pressure combustor shown in Fig. 1. The combustor is capable of operation at a pressure of up to 3 MPa, with a maximum airflow rate of 750 m_N3/h (0.3 kg/s) and exhaust gas temperatures up to 1850 K. The combustion air can be electrically preheated as high as 823 K. The

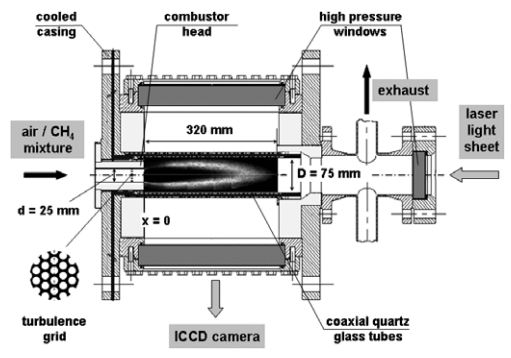

Fig. 1. Cross section through the combustor chamber.

combustor has a maximum thermal power of 400 kW. The cylindrical liner consists of two coaxial quartz glass tubes (inner quartz glass tube diameter $D = 75$ mm) which are convectively air-cooled. The combustor inlet diameter d is 25 mm. The flame is stabilized aerodynamically by the recirculation of hot flue gases induced by the sudden expansion geometry. The combustor is equipped with high-pressure windows to provide optical access to the entire combustor length.

The turbulence intensity u' and the integral length scale $L_{T,xx}$ in the inlet are varied by using different turbulence grids (defined by the hole diameter d_H and the blockage ratio b) or by mounting these grids at different axial positions xg within the combustor inlet section (see Fig. 1). In this study, perforated plates with hexagonally located holes, two different blockage ratios (50 and 65%) and two grid positions ($xg = 10$, 30) were used. The nomenclature of the grids is explained in Table 1. xg is the normalized grid position, defined as the axial distance from the grid position to the combustor inlet ($x = 0$ mm), divided by the grid hole diameter d_H. The grids used in the present experiments are listed in Table 1 together with the corresponding turbulence quantities at the combustor inlet measured with PIV in the isothermal flow field. The set-up used in the OH-PLIF measurements is described in detail in [18] and the flame detection methodology and the statistical analysis of the OH-PLIF images are explained in [16]. The spatial resolution of the PIV and OH-PLIF measurements were 0.6 and 0.5 mm, respectively. A post-processing algorithm based on the detection of the OH signal intensity gradient in the reaction zone leads to the flame front contour. For the statistical analysis 800 samples are summed up to create a 2D representation of the spatial flame front distribution (Fig. 2). Further analysis steps are applied to calculate quantities like the most probable flame front position at the centerline x_{MP} and the most probable flame front contour (MP) (white line in Fig. 2). x_{MP} is defined by the axial position of the maximum of the centerline profile extracted from the 2D image. The overall estimated error of the flame front position measurements x_{MP} is ±7%. The same centerline profile is used to define the flame brush thickness $\delta_T(x_{MP})$ which is defined as the distance between x_{MP} and the upstream axial location where the level of counts equals 10% of the centerline maximum. This definition is similar to the one proposed by Karpov and Severin [19]. They defined the flame brush thickness as the distance between the flame front leading edge and the surface of the "half-burning" which corresponds to the progress variable $\langle c \rangle = 0.5$ (which is roughly at x_{MP}, see Fig. 8). The estimated error of $\delta_T(x_{MP})$ is about ±10%.

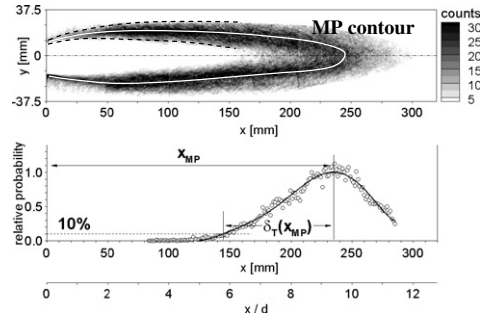

Fig. 2. Flame front detection and definition of x_{MP} and $\delta_T(x_{MP})$ (g365 × g10, 1 MPa, $\Phi = 0.43$, 673 K, 40 m/s).

The determination of the maximum OH-PLIF signal intensity in the y-direction at every x-position leads to the most probable flame front contour (MP). The virtual rotation of this contour line around the x-axis leads to a three-dimensional surface from which the turbulent flame speed can be calculated. In this paper, an alternative approach based on the progress variable ($\langle c \rangle = 0.05$), explained in Section 3.3, is used to calculate the flame front surface area A. The turbulent flame speed s_T was calculated according to Eq. (2) using an approach described in literature [15]. It is based on the mass continuity equation assuming that the density at the combustor inlet and in front of the flame front is the same.

$$s_T = u_{BULK} \frac{A_{inlet}}{A}. \tag{2}$$

Table 1
Turbulence characteristics at the combustor inlet for the different grids (d_H is the grid hole diameter, b is the blockage, xg is the dimensionless position of the grid in the inlet channel)

Grid case	d_H (mm)	b (%)	xg	u' (m/s)	$L_{T,xx}$ (mm)	Re_T
g350 × g30	3	50	30	2.03	3.1	430
g350 × g10	3	50	10	3.22	2.5	550
g450 × g10	4	50	10	3.93	2.7	720
g365 × g10	3	65	10	6.16	2.2	920
g465 × g10	4	65	10	5.29	3.4	1220

Re_T was calculated according Eq. (1) for 0.5 MPa, $\Phi = 0.5$.

In Eq. (2) A_{inlet} denotes the cross-sectional area of the combustor inlet and u_{BULK} is the inlet bulk velocity.

3. Results and discussion

3.1. Most probable flame front position

In Fig. 3 the dimensionless flame front positions (x_{MP}/d) are shown for a pressure range of 0.5–1.44 MPa at a preheating temperature of 673 K, an inlet velocity of 40 m/s and for the grid g350 × g10. The flame positions for pressures below 0.5 MPa are not presented because of the influence of the high combustor heat losses relative to the thermal power at these conditions. Over the entire pressure range the flame front position has a roughly constant value $x_{MP}/d \approx 7.4$ (±7%) indicating that the turbulent flame speed s_T is independent of pressure. This raises the question why s_T is independent of pressure whereas it is well known that the laminar flame speed of methane/air flames decreases with pressure approximately as $s_L \sim p^{-1/2}$ [17,20]. The explanation is that the decrease of s_L with pressure is compensated by an increase of the instantaneous flame front surface area. This increase is attributed to a higher flame front wrinkling induced by smaller turbulent eddies [12,21] with increasing pressure. It is known that the spectrum of turbulence broadens towards smaller scales as the pressure rises because the ratio of the integral length scale to the Kolmogorov scale η depends on pressure as $L_T/\eta \sim Re_T^{3/4} \sim p^{3/4}$. This broadening of the turbulence spectrum induces higher flame front wrinkling and in this way leads to an increase of the instantaneous flame front surface. Theoretically, a weak pressure dependence ($\sim p^{1/4}$) of s_T would be the result, however, this was not observed in the experiments.

Further analysis of Fig. 3 reveals that the flame front position (x_{MP}/d) strongly depends on the fuel concentration. Decreasing the fuel concentration results in an increase of x_{MP}/d by a factor of 2 from 5.5 at $\Phi = 0.56$ to 11.2 at $\Phi = 0.43$. This is caused by a decrease of the adiabatic flame temperature from 1860 to 1650 K leading to a drop of the global reaction rate and consequently to a lower s_T i.e. longer flames.

In Fig. 4 the dimensionless flame front positions (x_{MP}/d) are shown for variation of the preheating temperature (673–773 K) and the variation of the inlet bulk velocity (30–60 m/s) for flames at 0.5 MPa, $\Phi = 0.50$ and for the grid g365 × g10. As expected, the increase of the preheating temperature results in shorter flames. At $T = 673$ K the flame front is located at $x_{MP}/d \approx 5.4$ whereas at $T = 773$ K the flame front is shifted closer to the combustor inlet ($x_{MP}/d \approx 4.0$). This is due to the higher global reaction rate at higher temperatures leading to an increase of s_T i.e. to shorter flames. Increasing the inlet bulk velocity u_{BULK} from 30 to 60 m/s results in an increase of x_{MP}/d from 5.3 to 5.9 (see Fig. 4). This relatively small increase (11%) may be caused by higher flame front stretching due to higher velocity gradients in the shear layer which leads to more frequent local extinctions. Therefore the global reaction rate and consequently the turbulent flame speed are lower leading to longer flames.

The effect of turbulence on the flame front position is shown in Fig. 5, where x_{MP}/d is plotted as a function of the turbulent Reynolds number Re_T. The values of Re_T were calculated for every grid using the turbulence quantities (u', $L_{T,xx}$) at the combustor inlet (see Table 1). Each curve in Fig. 5 represents a grid variation with all other conditions being constant. The symbols correspond to the measurements at 0.5 MPa and equivalence ratios of $\Phi = 0.43$ (triangles), $\Phi = 0.50$ (circles) and $\Phi = 0.56$ (squares). It can be seen that all three curves show a similar trend. The flame front position shifts closer to the combustor inlet as Re_T increases indicating that the turbulent flame speed increases with Re_T. This increase of Re_T (u', $L_{T,xx}$) may lead to an intensified turbulent mass and heat transport of the hot flue gas from the recirculation zone into the fresh gas mixture.

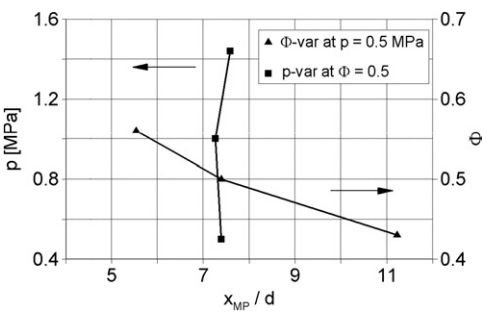

Fig. 3. Influence of pressure and equivalence ratio on x_{MP}/d (g350 × g10, 673 K, 40 m/s).

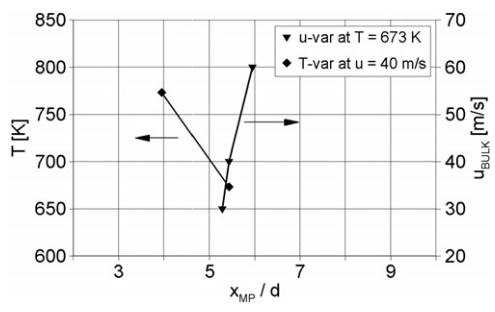

Fig. 4. Influence of preheating temperature and inlet bulk velocity on x_{MP}/d (g365 × g10, 0.5 MPa, $\Phi = 0.50$).

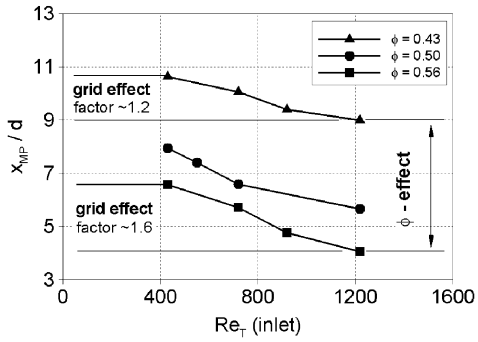

Fig. 5. Flame front position x_{MP}/d as a function of grid turbulence (Re_T) at $\Phi = 0.43$–0.56 and $p = 0.5$ MPa.

As a result the temperature of the fresh gas mixture and/or the radical concentration increase, both being known to increase the global reaction rate and therefore s_T. The turbulent Reynolds number could be varied from 430 to 1220 (factor of ~2.8) and the corresponding shortening of the flames (which slightly depends on the equivalence ratio) changes between a factor of roughly 1.6 at $\Phi = 0.56$ down to a factor of ~1.2 at $\Phi = 0.43$ (Fig. 5). This indicates that the influence of an enhanced turbulent transport in the vicinity of the combustor inlet has a bigger influence on the flame position for shorter flames i.e. at $\Phi = 0.56$. This can be explained by the fact that a relatively larger portion of the flame is exposed to the grid generated turbulence (which exists near the flame axis) when the flames are shorter (see also [16]).

3.2. Flame brush thickness

The development of the flame brush thickness δ_T/d along the axial distance from the combustor inlet x/d is presented in Fig. 6 for 673 K, 1 MPa, $\Phi = 0.43$, 40 m/s and the grid g365 × g10. The calculation method of δ_T is illustrated in Fig. 2 in which the black dashed lines mark the radial spreading of the flame brush. These lines correspond to the iso-contours of 10% of the maximum counts level of the entire image.

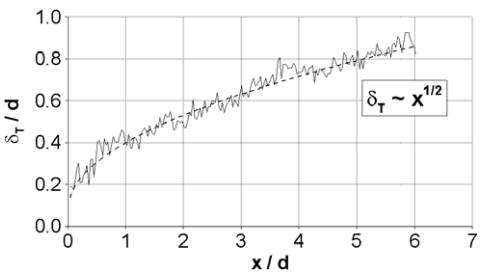

Fig. 6. Growth of the flame brush thickness along the axial distance from the combustor inlet (g365 × g10, 1 MPa, $\Phi = 0.43$, 673 K, 40 m/s).

The mean flame brush thickness grows with the axial distance from $\delta_T/d \approx 0.2$ at the inlet to $\delta_T/d \approx 0.8$ at $x/d = 6$. This growth is approximately proportional to the square-root of x (see the black dotted line in Fig. 6). The explanation is based on the conclusion that the mean flame brush thickness grows according to the turbulent diffusion law [2]. This law states that the spatial distribution of the turbulent fluctuations (e.g. u') and the flame front fluctuation must be the same. Based on the classical Taylor theory of turbulent diffusion (e.g. [22]) it can be shown that $\delta_T \sim x^{1/2}$. Such a close relation between spatial distribution of u' and δ_T has already been reported in previous literature [2,19,23,24].

In order to present the effects of operating conditions and turbulence on δ_T only the flame brush thickness $\delta_T(x_{MP})$ at the centerline is considered. In Fig. 7 the dimensionless flame brush thickness at the centerline $\delta_T(x_{MP})/d$ is plotted as a function of the most probable flame front position x_{MP}/d. All measurements were performed at 0.5 MPa. The points along the black solid line represent results for the grid g365 × g10. It can be seen that the decrease of the equivalence ratio ($\Phi = 0.56$–0.43 at 673 K and 40 m/s) results in a simultaneous increase of the flame front position (x_{MP}/d by a factor of ~2.7) and the flame brush thickness ($\delta_T(x_{MP})/d$ by a factor of ~2.4). This increase of $\delta_T(x_{MP})/d$ is mainly caused by the decrease of the stability (more susceptible to flow disturbances) of flames close to the lean blow-out limit [25]. Given the fact that the turbulence is not homogenous [26,27], the change of the flame brush thickness may be also caused by different local turbulence intensities at different flame front positions. In the present study, these effects cannot be quantified because the flow field was measured at non-reacting conditions. However, there seems to be a close relation between the flame

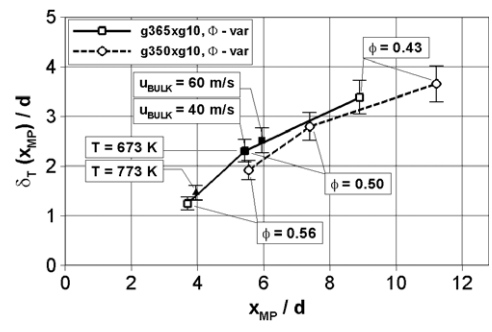

Fig. 7. Flame brush thickness $\delta_T(x_{MP})/d$ as a function of the flame front position x_{MP}/d for two grids and lambda, velocity and preheating temperature variations ($p = 0.5$ MPa).

brush thickness and the flame front position because all results of the grid g365 × g10 at different operating conditions (Φ, u_{BULK}, T) collapse on the same line (see Fig. 7). The increase of the inlet velocity from 40 to 60 m/s leads to a simultaneous increase of x_{MP}/d and $\delta_T(x_{MP})/d$. Even though the increase of both are relatively small (~10%) the trend is consistent with the previously discussed Φ variation results. Increasing the preheating temperature from 673 to 773 K results in the simultaneous decrease of x_{MP}/d and $\delta_T(x_{MP})/d$ (Fig. 7). This decrease of the flame brush thickness is most probably caused by a better flame stability at a higher temperature but it can also be an effect of a different u' at a different flame front position.

The effects of turbulence (grids g350 × g10, g365 × g10) on the flame brush thickness can be seen in the Φ variation series in Fig. 7. The flame front positions x_{MP}/d and the flame brush thicknesses $\delta_T(x_{MP})/d$ increase when the low-turbulence grid (g350 × g10) is used due to the x_{MP} shift already explained in Fig. 5. In Fig. 7 both grid results show the same close relation between x_{MP} and $\delta_T(x_{MP})$ which seems to be of a universal nature because the Φ variation results of both grids show a similar trend which resembles the square-root-like relation already observed in Fig. 6. No pressure effect on the flame brush thickness at the centerline $\delta_T(x_{MP})$ was observed.

3.3. Turbulent flame speed

One of the major problems in calculation of the turbulent flame speed s_T is the definition of the flame front surface area A used in Eq. (2). In this paper, an approach based on the progress variable, illustrated in Fig. 8 with quantities summarized in Table 2, is used to calculate the flame front surface area. The mean progress variable $\langle c\rangle = 0$ (black) refers to the fresh gas mixture and $\langle c\rangle = 1$ refers to completely burned gases (white). The black solid line corresponds to an iso-contour of $\langle c\rangle = 0.5$ which coincides with the most probable flame front contour (MP) represented by the white dashed line. The iso-contour of $\langle c\rangle$ =0.05, indicated by the solid white line, marks the leading edge of the flame front. The virtual rotation of this contour line ($\langle c\rangle = 0.05$) around the x-axis leads to a three-dimensional surface defined as the flame front surface area A. The turbulent flame speeds calculated according to Eq. (2) and A based on the progress variable approach are larger than the ones based on the most probable flame front contour (MP) [16] by a factor of ~1.5–2.0. The iso-surface $\langle c\rangle = 0.05$ represents the flame front leading edge where the temperature and density of the fresh gas mixture are regarded as unaffected by combustion. In this way the turbulent flame speed s_T and the laminar flame speed s_L consistently refer to the same conditions (unburned mixture). Thus, the calculation based on the flame front surface at the leading edge is a more appropriate way of defining s_T/s_L when comparing to literature and correlation data.

Table 2
Turbulence and flame characteristics (g350 × g30 at 0.5 MPa, 673 K, 40 m/s)

Φ	[ ]	0.56	0.50	0.43
s_L	[m/s]	0.35	0.24	0.12
δ_L	[mm]	0.043	0.062	0.125
$u'_{(at\ the\ inlet)}$	[m/s]	2.03	2.03	2.03
$L_{T,xx(\text{at the inlet})}$	[mm]	3.1	3.1	3.1
x_{MP}	[mm]	164.3	199.2	267.3
$\delta_T(x_{MP})$	[mm]	46.8	65.0	79.8
$s_{T(at\langle c\rangle=0.05)}$	[m/s]	2.25	1.73	1.12
$s_{T(Zimont)}$	[m/s]	1.81	1.50	1.06
$s_{T(Peters)}$	[m/s]	1.81	1.64	1.19

The experimental turbulent flame speed values, derived from the progress variable approach, are compared to correlation data (Zimont [8], Peters [7]) in Table 2. The s_T calculation was based on the turbulence parameters (u', $L_{T,xx}$) at the combustor inlet as commonly done in literature [13]. According to Table 2 the turbulent flame speed calculated at $\langle c\rangle = 0.05$ show a reasonable agreement with the data of both correlations (deviation ≈ 5–20%).

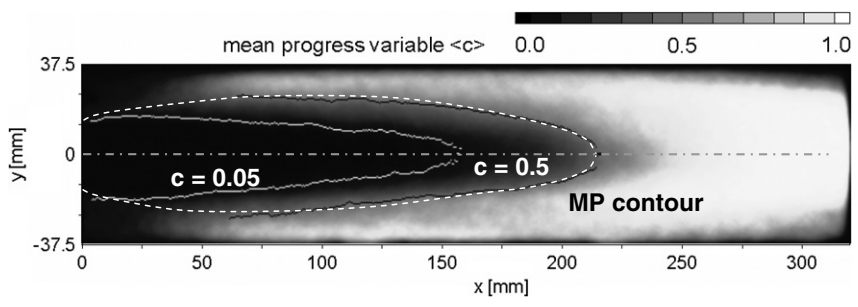

Fig. 8. Contour plot of the mean progress variable $\langle c\rangle$ (g365 × g10, 0.5 MPa, $\Phi = 0.43$, 40 m/s, 673 K).

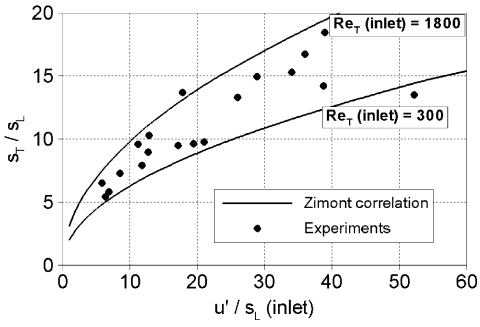

Fig. 9. Dimensionless turbulent flame speed s_T/s_L as a function of u'/s_L. Comparison of experimental and correlation data of Zimont [8].

A comparison of the experimental data with the Zimont correlation is presented in Fig. 9 in a dimensionless form $s_T/s_L = f(u'/s_L)$. The experimental data points represent the results at 673 K, $p = 0.35$–1 MPa, $\Phi = 0.43$–0.56, $u_{BULK} =$ 30–60 m/s and the two grids g365 × g10 and g350 × g30. The two black solid lines represent s_T/s_L from the Zimont correlation calculated for two different, constant turbulent Reynolds numbers corresponding to the experimental range mainly dependant on pressure ($Re_{T,min} = 300$, $Re_{T,max} = 1800$). Even though the scatter of the experimental data points is relatively large almost all the points fall within the range predicted by the Zimont correlation. Increasing u'/s_L results in a non-linear increase of s_T/s_L also known as the "bending effect." This is caused by additional effects like flame stretch and local extinction which have more influence at higher u'/s_L ratios. In general the agreement of the correlation data with the experimental results is satisfactory. However, an additional question should be addressed whether the s_T calculation using the turbulence parameters at the combustor inlet is suitable for such a configuration in which the flame stabilizes close to the shear layer. It is proposed here that the turbulence parameters averaged along the leading edge of the flame front ($\langle c \rangle = 0.05$) would be physically more meaningful. However, in this case the turbulence quantities would have to be measured under reacting conditions because it is expected that combustion influences the turbulence quantities close to the flame front location whereas the quantities at the combustor inlet should not be affected.

4. Conclusions

Turbulent flame speeds of lean premixed methane/air flames at high pressure have been measured and the effects of operating conditions and turbulence on flame front position and flame brush thickness have been investigated experimentally. A comparison of the measured turbulent flame speeds with existing correlations is given. The major results are the following:

1. No influence of pressure on the most probable flame front position, on the flame brush thickness, or on the turbulent flame speed was found in the pressure range of 0.5–1.44 MPa.
2. An increase of the turbulence intensity and the integral length scale at the combustor inlet leads to shorter flames i.e. to higher s_T. The maximum variation of the turbulent flame speed due to grid variation was 40%.
3. A close relation between the flame front position x_{MP} and the flame brush thickness $\delta_T(x_{MP})$ was observed ($\delta_T(x_{MP}) \sim x_{MP}^{1/2}$).
4. The comparison of the experimental turbulent flame speed data with the correlation of Zimont [8] shows satisfactory agreement.

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Comments

Andrei Lipatnikov, Chalmers University, Sweden. Your experimental data that indicate the independence of turbulent flame speed on pressure support the Zimont model, which predicts this independence for CH_4/air flames

Reply. This comment was already partly included ([4,7] in paper). Indeed, the results of the Zimont model agree well with the presented experimental results. They both show that the turbulent flame speed of methane/air flames is independent of pressure.

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Wolfgang Meier, German Aerospace Center DLR, Germany. From the OH PLIF image you showed, I expect that there is some local flame extinction. In that case, you get a bit of mixing of exhaust gas with fresh gas. Have you estimated the influence of this effect on your results?

Reply. We have observed some local flame extinction in our OH PLIF single shot images. However, for the statistical analysis this is not of a high relevance because local flame extinction occurred rather seldom. In principle, local flame extinction may lead to several counteracting effects. On one hand, flame front extinctions may lead to a decrease of the fuel consumption (i.e., to lower s_T). On the other hand, locally quenched flame fronts may be responsible for the creation of unburned gas pockets. These pockets, transported to the fresh gas mixture, may lead to an increase of the preheating temperature and/or to an increase of the radical concentration. As a consequence, this would increase the global reaction rate, i.e., s_T. To investigate and to quantify these effects it would be necessary to measure, for example, the local gas temperature which was not possible in the present study.